ISYE 6740 Summer 2026
Homework 4
(100 points)

Student Name Here

Provided Data and Submission Requirements:

- All results, explanations, and math must be included in your PDF submission. Handwritten work is not accepted per the syllabus. Screenshots of work is also not accepted.
- If applicable, questions marked with **GS** must be submitted to Gradescope. Failure to pass gradescope tests will result in penalties to the points for that question.

This assignment does not have any gradescope requirements
Assignment Data:

- Q1: Comparing Classifiers: Divorce Classification / Prediction [marriage.csv]
- Q2: Comparing multi-class classifiers for MNIST Datasets [mnist_10digits.mat, fashion-mnist_train.csv, fashion-mnist_test.csv]

1. Comparing classifiers: Divorce classification/prediction (20 points)

In lectures, we learn different classifiers. This question is compare them on two datasets. Python users, please feel free to use Scikit-learn, which is a commonly-used and powerful Python library with various machine learning tools. But you can also use other similar libraries in other languages of your choice to perform the tasks.

This dataset is about participants who completed the personal information form and a divorce predictors scale. The data is a modified version of the publicly available at `https://archive.ics.uci.edu/dataset/539/divorce+predictors+data+set` (by injecting noise so you will not get the exactly same results as on UCI website). The dataset **marriage.csv** is contained in the homework folder. There are 170 participants and 54 attributes (or predictor variables) that are all real-valued. The last column of the CSV file is label y (1 means "divorce", 0 means "no divorce"). Each column is for one feature (predictor variable), and each row is a sample (participant). A detailed explanation for each feature (predictor variable) can be found at the website link above. Our goal is to build a classifier using training data, such that given a test sample, we can classify (or essentially predict) whether its label is 0 ("no divorce") or 1 ("divorce").

We are going to compare the following classifiers (**Naive Bayes, Logistic Regression, and KNN**). Use the first 80% data for training and the remaining 20% for testing. If you use scikit-learn you can use train_test_split to split the dataset.

Remark: Please note that, here, for Naive Bayes, this means that we have to estimate the variance for each individual feature from training data. When estimating the variance, if the variance is zero to close to zero (meaning that there is very little variability in the feature), you can set the variance to be a small number, e.g., $\epsilon = 10^{-3}$. We do not want to have include zero or nearly variance in Naive Bayes. This tip holds for both Part One and Part Two of this question.

1. (10 points) Report testing accuracy for each of the three classifiers. Comment on their performance: which performs the best and make a guess why they perform the best in this setting.

2. (10 points) Now perform PCA to project the data into two-dimensional space. Build the classifiers (**Naive Bayes, Logistic Regression, and KNN**) using the two-dimensional PCA results. Plot the data points and decision boundary of each classifier in the two-dimensional space. Comment on the difference between the decision boundary for the three classifiers. Please clearly represent the data points with different labels using different colors.

2. Comparing multi-class classifiers for MNIST Datasets. (15 points)

This question will compare different classifiers and their performance for multi-class classifications on the complete MNIST dataset, looking at both `MNIST_Digits` and `MNIST_Fashion`. Use the data files provided in the homework folder. Both MNIST datasets have 60,000 training examples and 10,000 test examples. We will compare **KNN, logistic regression, SVM, kernel SVM, and neural networks**.

- Use 6740 as your random_state random seed
- You should standardize the data before training your classifiers by mapping the range from [0,255] to [0,1].
- You will run your models on both MNIST_DIGITS and MNIST_FASHION. The shape of your data after loading should be as follows:
  - xtrain (60000, 784)
  - xtest (10000, 784)
  - ytrain (60000,)
  - ytest (10000,)
- load `MNIST_DIGITS` using `scipy.io.loadmat`
- Utilize the following models from sklearn:

- LogisticRegression
- KNeighborsClassifier : Adjust K to obtain a reasonable result. Feel free to use any reasonable tuning methodology, only report your best K
- MLPClassifier : Use hidden_layer_sizes = (20, 10)
- SVC : Use this for both SVM and Kernel SVM. Randomly downsample the training data to size $m = 5000$ for efficiency. For kernel SVM, use radial basis function kernel (rbf).

Train the classifiers on the training dataset and evaluate them on the test dataset.

1. (10 points) Report the precision, recall, and F-1 score for each of the classifiers. The classification metrics should be reported for each class in the dataset. Provide these results for both the digits and fashion datasets.

2. (5 points) Discuss the performance of the classifiers. Include explanations both on comparing the 5 classifiers against each other for the individual datasets, and some insight into performance on the different datasets. Which model performed best on each dataset, and why might some models perform better than others?

3. SVM (25 points).

1. (5 points) Explain why can we set the margin $c = 1$ to derive the SVM formulation? Justify using a mathematical proof.

2. (5 points) Using Lagrangian dual formulation, show that the weight vector can be represented as

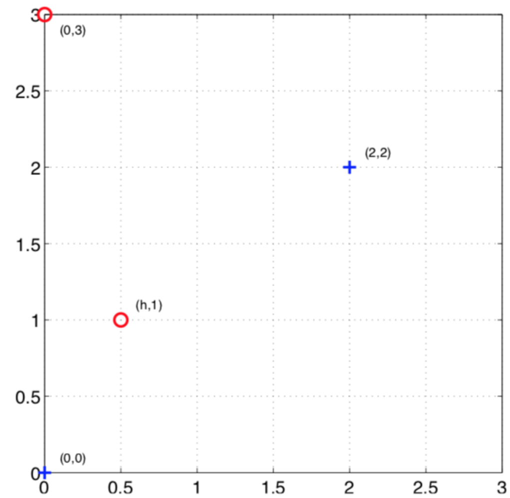
$$w = \sum_{i=1}^{n} \alpha_i y_i x_i.$$

   where $\alpha_i \geq 0$ are the dual variables. What does this imply in terms of how to relate data to w?

3. (5 points) Explain why only the data points on the "margin" will contribute to the sum above, i.e., playing a role in defining w. Hint: use the Lagrangian multiplier derivation and KKT condition we discussed in class.

4. Simple SVM by hand.

   Suppose we only have four training examples in two dimensions as shown in Fig. The positive samples at $x_1 = (0, 0)$, $x_2 = (2, 2)$ and negative samples at $x_3 = (h, 1)$ and $x_4 = (0, 3)$.



(a) (5 points) For what range of parameter $h > 0$, the training points are still linearly separable? Be sure to consider all possible ranges.

(b) (5 points) Does the orientation of the maximum margin decision boundary change as h changes, when the points are separable? Please explain your conclusion.

4. Neural networks and backpropagation (20 points)

Consider a simple two-layer network in the lecture slides. Given m training data (x^i, y^i), $i = 1, \ldots, m$, the cost function used to training the neural networks

$$\ell(w, \alpha, \beta) = \sum_{i=1}^{m} (y^i - \sigma(w^T z^i))^2$$

where $\sigma(x) = 1/(1 + e^{-x})$ is the sigmoid function, z^i is a two-dimensional vector such that $z_1^i = \sigma(\alpha^T x^i)$, and $z_2^i = \sigma(\beta^T x^i)$.
Be sure to show all steps of your proofs.

1. (10 points) Show that the gradient is given by

$$\frac{\partial \ell(w, \alpha, \beta)}{\partial w} = -\sum_{i=1}^{m} 2(y^i - \sigma(u^i))\sigma(u^i)(1 - \sigma(u^i))z^i,$$

   where $u^i = w^T z^i$.

2. (10 points) Also, show the gradient of $\ell(w, \alpha, \beta)$ with respect to α and β and write down their expression.

5. Feature selection and change-point detection. (20 points)

1. (10 points) Consider the mutual information-based feature selection. Suppose we have the following table (the entries in the table indicate counts) for the spam versus and non-spam emails:

	"prize" = 1	"prize" = 0
"spam" = 1	145	35
"spam" = 0	1540	16740

	"hello" = 1	"hello" = 0
"spam" = 1	190	55
"spam" = 0	12200	6740

   Given the two tables above, calculate the mutual information for the two keywords, "prize" and "hello" respectively. Which keyword is more informative for deciding whether or not the email is spam? If any tools are used for your calculation, you must still show your mathematical steps in your report and include code/files used for your calculations.

2. (10 points) Given two distributions, $f_0 = \mathcal{N}(0, 1)$, $f_1 = \mathcal{N}(0, 1.25)$, derive what should be the CUSUM statistic (i.e., show the mathematical CUSUM detection statistic specific to these distributions). Plot the CUSUM statistic for a sequence of 150 randomly generated i.i.d. (independent and identically distributed) samples, $x_1, \ldots, x_{100}$ according to f_0 and $x_{101}, \ldots, x_{150}$ according to f_1. Using your plot, please provide a reasonable estimation of where a change may be detected.
   Note: Be certain to set your random seed to 6740 to ensure reproducible results. Failure to do so will result in point deductions